

README

Overview

This replication package is structured as follows:

- Folder **Code_Taxes**: contains the code needed to replicate the results from Section 2.1 and the Online Appendix S3 on the paper – *The Effect of Corporate Taxes on Investment and Entrepreneurship* – by Djankov et al., (2010a). For abbreviation purposes we refer to this paper further in this document as the **Taxes** paper. The replicator should expect to run this part of the code for about 4 hours. For example, Line 1 of Table 1 Panel A takes 32 minutes 20 seconds; Line 1 of Table 1 Panel B takes 16 minutes 22 seconds.
- Folder **Code_Tariffs**: contains the code needed to replicate the results from Section 2.2 and the Online Appendix S3 on the paper – *The Structure of Tariffs and Long-term Growth* – by Nunn and Treffer (2010a). For abbreviation purposes we refer to this paper further in this document as the **Tariffs** paper. The replicator should expect to run this part of the code for about 50 hours. For example, Table 2 Panel A takes 28 minutes 36 seconds; Table S3.3 Panel A takes 7 hours 58 minutes.
- Folder **Code_FoxNews**: contains the code needed to replicate the results from Section 3.1 and the Online Appendix S3 on the paper – *The Fox News Effect: Media Bias and Voting* – by DellaVigna and Kaplan (2007a). For abbreviation purposes we refer to this paper further in this document as the **Fox News** paper. The replicator should expect to run this part of the code for about 65 minutes. Each table of this application takes about 5-10 minutes.
- Folder **Code_TeacherTraining**: contains the code needed to replicate the results from Section 3.2 and the Online Appendix S3 on the paper – *Does Teacher Training Actually Work? Evidence from a Large-scale Randomized Evaluation of a National Teacher Training Program* – by Loyalka et al. (2019a). For abbreviation purposes we refer to this paper further in this document as the **Teacher Training** paper. The replicator should expect to run this part of the code for about 7 hours, after which all tables referring to this application are obtained.
- Folder **Data**: contains a copy of the data sets corresponding to all four papers above. See details on the Data Availability and Provenance below. All data sets are publicly available either from the journal where the paper is published or from the authors' web page. See details below. We also provide in this folder the data in .csv format.
- Folder **MonteCarlo**: contains the code needed to replicate the results from the Online Appendix S4 on the Monte Carlo simulations. For example, the second panel of Table S4.17 takes 22 minutes and 59 seconds; the second

panel for Table S4.22 takes 38 minutes and 58 seconds.

Data Availability and Provenance Statements

This section provides information on the Data Availability, Provenance and Rights. Note that variables with long-form meaningful names and their values for each data file are described in the Stata .dta files provided in the Data folder, except for the Teacher Training paper, for which we provide the file *TeacherTraining_VariablesDescription.pdf* in the Data folder for this purpose. For further information one can also refer to a detailed variables description in the revisited papers.

Statement about Rights

The authors of the manuscript certify that they have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

All data are publicly available, see table below for the sources. A copy of the data is also provided as part of this archive in both original .dta format as well as open-source .csv format.

Details on each Data File

Data file	Source	Notes	Provided
data_taxes.dta	Journal's webpage	The data (Djankov et al., 2010b) for the Taxes paper is publicly available on the American Economic Journal: Macroeconomics webpage. It can be downloaded here .	Yes
data_tariffs.dta	Journal's webpage	The data (Nunn and Treffer, 2010b) for the Tariffs paper is publicly available on the American Economic Journal: Macroeconomics webpage. It can be downloaded here .	Yes

Data file	Source	Notes	Provided
data_fox_news.dta	Stefano DellaVigna's webpage	The data (DellaVigna and Kaplan, 2007b) for the Fox News paper is publicly available on Stefano DellaVigna's webpage. It can be downloaded here .	Yes
data_teacher_training.dta	Journal's webpage	The data (Loyalka et al., 2019b) for the Teacher Training paper is available on the American Economic Journal: Applied Economics webpage. It can be downloaded here .	Yes

Dataset list

Computational requirements

Software Requirements

- Code was last run with **R** version: R 4.2.1 GUI 1.79 High Sierra build (8095). **Rstudio version:** RStudio 2022.07.2+576 "Spotted Wakerobin" Release.

Controlled Randomness

- **Taxes**
Random seed is set at line 42 of programs `taxes_inv_03_05_A.R`, `taxes_fdi_A.R`, `taxes_business_density` and `taxes_entryrate`.
- **Tariffs**
Random seed is set at line 41 of programs `dml_tariffs_T2.R` and `dml_tariffs_appendix_tables.R`.
- **Fox News**
Random seed is set at line 17 and 15 of program `causal_forest_ttest30_clust.R` and `causal_forest_ttest30_fe.R`, respectively.
- **Teacher Training**
Random seed is set at line 84 of program `el_teachertraining_rep28.R`

– **Monte Carlo**

Random seed is set at line 53 of program `main.r` and at line 53 of program `MonteCarloDML.r`.

Memory and Runtime Requirements

Summary Approximate total time needed to reproduce the analyses on a standard (2022) desktop machine: 62h for the empirical part and 36h for the Monte Carlo part. See details above on examples of computation times for specific tables in each application/section.

Details The code was last run on a **Quad-Core Intel Core i7 laptop with MacOS version 10.15.6**.

Portions of the code (Tables S4.19, S4.20, S4.23 and S4.24 of the Monte Carlo simulations) were run on a **Microsoft Azure** virtual machine (VM). Details of this VM are provided in the Annex below.

Description of programs/code

– **Taxes**

`Code_Taxes/taxes/taxes_inv_03_05_A.R` generates Panel A of Table 1 in the main text.

`Code_Taxes/taxes/taxes_fdi_A.R` generates Panel B of Table 1 in the main text.

`Code_Taxes/taxes/taxes_business_density` generates Panel C of Table 1 in the main text.

`Code_Taxes/taxes/taxes_entryrate` generates Panel D of Table 1 in the main text and Table S3.1 in the online appendix.

`Code_Taxes/taxes/taxes_inv_03_05_A_cf.R` generates column 1 of Table S3.2 in the online appendix.

`Code_Taxes/taxes/taxes_fdi_A_cf.R` generates column 2 of Table S3.2 in the online appendix.

`Code_Taxes/taxes/taxes_business_density_B_cf.R` generates column 3 of Table S3.2 in the online appendix.

`Code_Taxes/taxes/taxes_entryrate_B_cf.R` generates column 4 of Table S3.2 in the online appendix.

– **Tariffs**

`Code_Tariffs/dml_tariffs_T2.R` generates Table 2 from the main text and `Code_Tariffs/dml_tariffs_appendix_tables.R` generates Tables S3.3 and S3.4 from the online appendix.

`Code_Tariffs/data_tariffs_tabel_column.R` contains the lines of code to set the correct data that needs to be copy-pasted into lines 51-61 in `Code_Tariffs/dml_tariffs_T4.R`, depending on what table and panel you want to generate. `Code_Tariffs/GRF_tariffs_T4.Rmd` generates table S3.5.

– **Fox News**

`Code_FoxNews/causal_forest_ttest30_fe.R` generates column 1 of Table 3 and Panel A of Table 4 from the main text. It also generates Tables S3.7, S3.9, Column 1 of Table S3.6, Column 1 of Table S3.11 and Panel A of Table S3.12 from the online appendix.

`Code_FoxNews/causal_forest_ttest30_clust.R` generates column 2 of Table 3 and Panel B of Table 4 from the main text. It also generates Tables S3.8, S3.10, Column 2 of Table S3.6, Column 2 of Table S3.11 and Panel B of Table S3.12 from the online appendix.

– **Teacher Training**

`Code_TeacherTraining/el_teachertraining_rep28.R` generates Table 5, Figure 1 and Table 6 from the main text. It also generates Tables S3.13, S3.14, Table S3.15 and Table S3.16 from the online appendix.

– **Monte Carlo**

`MonteCarlo/main.r` generates Table S4.17 to S4.24 of the online appendix. Table outputs for all tables are exported to output files with relevant names such as `2PLR_n100_sim100_k90_linear_normal.txt`

R Packages

The user might need to install the following R packages: `foreign`, `tidyverse`, `quantreg`, `gbm`, `glmnet`, `MASS`, `rpart`, `doParallel`, `sandwich`, `randomForest`, `nnet`, `matrixStats`, `xtable`, `readstata13`, `car`, `lfe`, `caret`, `foreach`, `multcomp`, `cowplot`, `fastDummies`, `devtools`, `DiagrammeR`, `DiagrammeRsvg`, `plyr`, `knitr`, `doBy`, `mnormt`, `iterators`, `parallel`, `hdm`, `neuralnet`, `quadprog`, `ivmodel`, `Hmisc`, `dummy`, `gtools`, `tictoc`, `grf` (version 1.2.0 from CRAN). We also used the MRAN RStudio package manager, dated to 2022-01-03.

Instructions to Replicators

We detail below the instructions for replication for the different parts of the paper.

The code is written in R. We suggest to use Docker Hub image of RStudio server, *which includes all the necessary R packages pre-installed*. Detailed instructions on how to use this tool can be found in the file *"docker_instructions.html"*. Alternatively, the code can be also run on your local machine.

– **Taxes**

- * Download the data and code files referenced above. If necessary install requisite R packages listed at the beginning of the code files.

- * Edit `Code_Taxes/taxes/taxes_inv_03_05_A.R` to adjust the path to the working directory in line 35.
 - * Run `Code_Taxes/taxes/taxes_inv_03_05_A.R`. The results will be found as text files as outlined in the table below.
 - * Proceed similarly for `Code_Taxes/taxes/taxes_fdi_A.R`, `Code_Taxes/taxes/taxes_business_density_B.R`, `Code_Taxes/taxes/taxes_entryrate_B.R`, `taxes_inv_03_05_A_cf`, `Code_Taxes/taxes/taxes_fdi_A_cf.R`, `Code_Taxes/taxes/taxes_business_density_B_cf.R` and `Code_Taxes/taxes/taxes_entryrate_B_cf.R`.
- **Tariffs**
- * Open `Code_Tariffs/data_tariffs_table_column.R` and copy the lines corresponding to the tables and columns you wish to replicate. The comments in `data_tariffs_table_column.R` indicate which input variables have to be used for the different tables and columns.
 - * To replicate Table 2: Open `Code_Tariffs/dml_tariffs_T2.R` and change lines 51-61 to your copied lines. Set your working directory on line 37. Run `Code_Tariffs/dml_tariffs_T2.R`. The results will be found as text files as outlined in the table below.
 - * To replicate Tables S3.3 and S3.4 from the online appendix: Open `Code_Tariffs/dml_tariffs_appendix_tables.R` and change lines 51-61 to your copied lines. Set your working directory on line 37. Run `Code_Tariffs/dml_tariffs_appendix_tables.R`. The results will be found as text files as outlined in the table below.
 - * Finally, to replicate Table S3.5, open the file `GRF_tariffs_Table4.Rmd`, adjust the path to the working directory in line 14 and run. The results will be found as text files as outlined in the table below.
- **Fox News**
- * Download the data and code files referenced above. If necessary install requisite R packages listed at the beginning of the code files.
 - * Edit `Code_FoxNews/causal_forest_ttest30_fe.R` and `Code_FoxNews/causal_forest_ttest30_clust.R` to adjust the path to the working directory in line 14 and 16, respectively.
 - * Run `Code_FoxNews/causal_forest_ttest30_fe.R` and `Code_FoxNews/causal_forest_ttest30_clust.R`. The results will be found as text files as outlined in the table below.
- **Teacher Training**
- * Download the data and code files referenced above. If necessary install requisite R packages listed at the beginning of the code file.
 - * Edit `Code_TeacherTraining/el_teachertraining_rep28.R` to

adjust the path to the working directory in line 7.

- * Run `Code_TeacherTraining/el_teachertraining_rep28.R`, The results will be found as text files as outlined in the table below.

– **Monte Carlo**

- * Open `MonteCarlo/main.R` and change line 3 to set your own directory.
- * Switch between lines 42 and 43 for linear and nonlinear DGPs, respectively.
- * Switch between lines 45 and 46 for normal and uniform error and covariate distributions, respectively.
- * Set k , and n on lines 28 and 29 to the values of the tables that you want to replicate.
- * Run `MonteCarlo/main.R`. The results will be found as text files as outlined in the table below.

List of tables and programs

Details

The provided code reproduces all tables in the paper and in the online appendix.

Figure/ Table #	Program	Line Number	Output File
Taxes			
Table 1	taxes_inv_03_05_A.R	all	TX_T1_PA_LX.txt, X=1,2,3
	taxes_fdi_A.R		TX_T1_PB_LX.txt, X=1,2,3
	taxes_business_density_B.R		TX_T1_PC_LX.txt, X=1,2,3
Table S3.1/ Figure S3.1	taxes_entryrate_B.R	112-157	TX_T1_PD_LX.txt, X=1,2,3
	taxes_inv_03_05_A_cf.R		TX_S1.txt, TX_S1.1.png, TX_S1.2.png
Table S3.2	taxes_fdi_A_cf.R	all	TX_S2_C1.txt
	taxes_business_density_B_cf.R		TX_S2_C2.txt
	taxes_entryrate_B_cf.R		TX_S2_C3.txt TX_S2_C4.txt
Tariffs			
Table 2	dml_tariffs_T2.R	all	TA_T2_PA.txt, TA_T2_PA_boosting.txt, TA_T2_PB.txt, TA_T2_PB_boosting.txt, TA_T2_PC.txt, TA_T2_PC_boosting.txt
Table S3.3	dml_tariffs_appendix_tables.R	all	TA_S3_PA.txt TA_S3_PB.txt TA_S3_PC.txt
Table S3.4	dml_tariffs_appendix_tables.R	all	TA_S4_PA.txt TA_S4_PB.txt TA_S4_PC.txt
Table S3.5	GRF_tariffs_Table4.Rmd	all	TA_S5.txt
Fox News			
Table 3	causal_forest_ttest30_fe.R	116	FN_T3_C1.txt
	causal_forest_ttest30_clust.R	108	FN_T3_C2.txt
Table 4	causal_forest_ttest30_fe.R	163	FN_T4_PA_&_S7.txt
	causal_forest_ttest30_clust.R	156	FN_T4_PB_&_S8.txt
Table S3.6	causal_forest_ttest30_fe.R	169	FN_S6_C1.txt
	causal_forest_ttest30_clust.R	162	FN_S6_C2.txt
Table S3.7	causal_forest_ttest30_fe.R	163	FN_T4_PA_&_S7.txt
Table S3.8	causal_forest_ttest30_clust.R	156	FN_T4_PB_&_S8.txt
Table S3.9	causal_forest_ttest30_fe.R	129	FN_S9.txt
Table S3.10	causal_forest_ttest30_clust.R	122	FN_S10.txt
Table S3.11	causal_forest_ttest30_fe.R	231	FN_S11_C1.txt
	causal_forest_ttest30_clust.R	224	FN_S11_C1.txt
Table S3.12	causal_forest_ttest30_fe.R	268	FN_S12_PA.txt
	causal_forest_ttest30_clust.R	260	FN_S12_PB.txt
Teacher Training			
Table 5	el_teachertraining_rep28.R	663	EL_FN_BLP20- range-best-2-2-100.txt
Figure 1	el_teachertraining_rep28.R	668	EL_FN_plot_best-range -best-2-2-100-z_endline.pdf
Table 6	el_teachertraining_rep28.R	666	EL_FN_CLAN20- range-best-2-2-100.txt
Table S3.13	el_teachertraining_rep28.R	665	EL_FN_BEST20- range-best-2-2-100.txt
Table S3.14	el_teachertraining_rep28.R	664	EL_FN_GATES20- range-best-2-2-100.txt
Table S3.15	el_teachertraining_rep28.R	666	EL_FN_CLAN20- range-best-2-2-100.txt
Table S3.16	el_teachertraining_rep28.R	500	TT_S16.txt
Monte Carlo			
Table S4.17	main.R	all	2PLR_n100_sim100_k20_linear_normal.txt 2PLR_n100_sim100_k50_linear_normal.txt 2PLR_n100_sim100_k90_linear_normal.txt
Table S4.18 to S4.24	main.R	all	Similar output file names as for Table S4.17 depending on the MC setting

References

- DellaVigna, S. and E. Kaplan (2007a). The fox news effect: Media bias and voting. *The Quarterly Journal of Economics* 122, 1187–1234.
- DellaVigna, S. and E. Kaplan (2007b). The fox news effect: Media bias and voting. *The Quarterly Journal of Economics*. Data available at <https://eml.berkeley.edu/~sdellavi/data/foxnewsdata.shtml>.
- Djankov, S., T. Ganser, C. McLiesh, R. Ramalho, and A. Shleifer (2010a). The effect of corporate taxes on investment and entrepreneurship. *American Economic Journal: Macroeconomics* 2, 31–64.
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- Nunn, N. and D. Trefler (2010a). The structure of tariffs and long-term growth. *American Economic Journal: Macroeconomics* 2, 158–94.
- Nunn, N. and D. Trefler (2010b). The structure of tariffs and long-term growth. *American Economic Journal: Macroeconomics*. Data deposited at ICPSR, <https://www.openicpsr.org/openicpsr/project/114183/version/V1/view>

Annex

Details for Microsoft Azure environment, Monte Carlo Simulations (for $n = 600, k = 120$):

- Region: East US 2
- Type: **Standard F4s v2**
- Number of vCPUs: 4
- Possible CPU generations (at the time of running the user can get assigned to):
 - 3rd Generation Intel Xeon Platinum 8370C (Ice Lake)
 - Intel Xeon Platinum 8272CL (Cascade Lake)
 - Intel Xeon Platinum 8168 (Skylake)
- Memory: 8 GB
- Operating system: Linux Ubuntu 20.04

Details for Microsoft Azure environment, Monte Carlo Simulations (for $n = 600, k \in \{300, 540\}$):

- Region: East US
- Type: **Standard HB120rs v2**
- Number of vCPUs: 120
- Possible CPU generations (at the time of running the user can get assigned to):
 - AMD EPYC 7V12
 - AMD EPYC 7742
- Memory: 456 GB
- Operating system: Linux Ubuntu 20.04